Diagnosis

Clinical

The staging of syphilis is clinical, with the chronology beginning from the onset of the chancre. The stages often overlap. Secondary syphilis develops in approximately one-third of untreated individuals, and tertiary syphilis occurs in about 10%. Patients are most infectious during the first year of infection—particularly during the primary and secondary stages—mainly through sexual contact, though rare transmission via social contact can occur. Later transmission typically occurs through vertical (mother-to-child) routes or via tissue and organ donation.

Incubation period: 10–90 days between exposure (usually sexual) and the appearance of the chancre.

Primary Syphilis

Characterized by the development of a painless, indurated ulcer known as a **chancre**, typically accompanied by regional lymphadenopathy. The chancre is usually: - Superficial - Single - Painless - Indurated - With a clean base that exudes clear serous fluid

It most commonly appears in the anogenital region. It does **not** present as a blistering lesion. However, atypical presentations are frequent—lesions may be multiple, painful, deep, or clinically indistinguishable from herpes simplex virus ulcers.

In women and men who have sex with men (MSM), chancres may be difficult to detect due to their location (e.g., cervical, vaginal, rectal). Any anogenital ulcer should be presumed syphilitic until proven otherwise.

Initial testing may not definitively exclude syphilis. In cases of high clinical suspicion with negative initial serology, repeat serological testing at 1, 2, and 6 weeks is recommended. However, delaying treatment is potentially dangerous, especially in populations with poor follow-up adherence.

Secondary Syphilis

Occurs within the first year of infection (may recur into the second year) due to hematogenous dissemination of *Treponema pallidum*.
Present in 90% of cases with: - Non-itchy skin rash (roseolar rash appearing 2–3 months after chancre; later papular syphilids) - Mucocutaneous lesions

Systemic manifestations may include: - Fever - Generalized lymphadenopathy - Hepatitis - Splenomegaly - Periostitis - Arthritis - Glomerulonephritis

Neurological involvement can occur and should be considered **early neurosyphilis**, including: - Meningitis - Cranial nerve palsies - Ocular manifestations (e.g., uveitis, retinitis, papilledema) - Auditory involvement (e.g., otitis) - Meningovascular syphilis (e.g., stroke, myelitis)

Latent Syphilis

Defined by positive serological tests for syphilis in the absence of clinical signs of active infection.

Arbitrarily classified as: - **Early latent syphilis**: infection acquired within the past year. This includes individuals who: - Had a negative syphilis serology within the past year - Show a fourfold (two dilutions) or greater decline in non-treponemal antibody titers - Have clear evidence of recent exposure (clinical disease in self or partner) - **Late latent syphilis** (or latent syphilis of unknown duration): infection present for more than one year

Tertiary Syphilis

Occurs in a subset of untreated individuals, typically years after initial infection.

Manifestations include: - **Gummatous syphilis**: granulomatous lesions presenting as nodules, plaques, or ulcers affecting skin, mucosa, or internal organs - **Late neurosyphilis**: - Meningitis - Cranial nerve dysfunction - Meningovascular syphilis (stroke, myelitis) - Parenchymatous neurosyphilis (general paresis, tabes dorsalis) - **Cardiovascular syphilis**: - Aortic regurgitation - Stenosis of coronary ostia - Thoracic aortic aneurysm

Neurologic Syphilis

Can occur at any stage: - **Early**: during secondary syphilis - **Late**: in tertiary syphilis
Includes meningitis, cranial nerve deficits, and meningovascular or parenchymal central nervous system involvement.

Laboratory

Demonstration of T. pallidum

Direct detection methods provide definitive diagnosis of syphilis.

- **Darkfield microscopy (DFM)**:
  Useful for examining fluid from chancres or erosive mucocutaneous lesions. Provides immediate results but requires expertise. Limitations include:
- Subjective interpretation
- Risk of false positives and frequent false negatives
- Not reliable for oral or rectal lesions due to presence of non-pathogenic treponemes

- **Polymerase chain reaction (PCR)**:
  Preferred method for confirming *T. pallidum* in clinical samples, especially:

- Oral, rectal, or genital lesions
- Cerebrospinal fluid (CSF)
- Tissue specimens
  Less sensitive in blood samples
  Advantages over darkfield: higher specificity, especially in sites colonized by commensal treponemes